# YAO XIANG

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NUS Computer Engineering undergraduate (Minor in AI) combining autonomous multi-model agent systems, real-time distributed backends, and accelerated edge AI for robotics. Experienced leading technical teams, architecting fault-tolerant async pipelines, and deploying mission-critical systems solo on-site.

## EDUCATION & COMPETITIVE HACKATHONS

Verified Denominators

**National University of Singapore (NUS)** — Aug 2024 – Aug 2028 (Projected)

Bachelor of Engineering in Computer Engineering • Specialization: **Advanced Electronics & IoT** • Minor: **Artificial Intelligence**

- **Singapore Defence Tech Hackathon (SDTH)**: Top 6 / 800 participants (LOTR track), Sep 2026 — Top 0.75% — **Marsham Edge**
- **DSTA BrainHack CODE_EXP 2026**: Open Finalist (4,500+ students across all BrainHack tracks, Jun 2026) — **Kampung Kaki**
- **Vibe For Good 2026**: Semi-Finalist (AI venture co-builder & creator platform, Sep 2026) — **VibeCheck**

**2026 Hackathons (11 Competitions): Singapore Defence Tech Hackathon** (Finalist, Top 6, LOTR track) • **DSTA BrainHack CODE_EXP** (Open Finalist) • **Vibe For Good** (Semi-Finalist) • NUS × Synapxe × IMDA AI Innovation Challenge • AI Engineer Hackathon • TinyFish SG Hackathon • Hack4Good • VibeWithSG × Duality AI • Tencent Cloud Age Well Social Good Challenge • Huawei Tech4City • SUP Buildathon

## WORK EXPERIENCE

**KABAM Robotics** – Software Robotics Intern (Solo On-Site Deployment) — May 2026 – Present (until Dec 2026)

**Stack:** `Jetson Orin AGX, TensorRT, SAM3, YOLO11, FFmpeg, ROS 2, LiDAR SLAM, Nav2 MPPI, CAN bus, MWAN3, Docker`

- **Sole On-Site Deployment:** Conducted first ever SG field PoC as solo on-site robotics engineer for KABAM's latest patrol robot model of Jul 2026; diagnosed LiDAR SLAM drift, Nav2 MPPI trajectory tracking, autonomous docking, and MWAN3 multi-WAN networking alongside remote engineers.
- **Edge CV Acceleration:** Optimized Segment Anything Model (SAM3) vision-language inference on Jetson Orin AGX with TensorRT; piped real-time RTSP/FFmpeg video streams with YOLO11 perception overlays.
- **Diagnostic & Mission Tooling:** Shipped cross-platform desktop ROS 2 diagnostics and mobile mission-control tools featuring hardware emergency-stop overrides, CAN bus telemetry decoders, and live LiDAR/occupancy grid streaming.

**National University of Singapore** – Teaching Assistant (CS2113 Software Engineering & OOP) — Jan 2026 – May 2026

- Mentored undergraduate teams on Java object-oriented architecture, JUnit test-driven design, Git collaboration, and CI/CD pipelines.

## FEATURED ENGINEERING PROJECTS

**Marsham Edge** – Counter-Drone Decision Console & Simulation (Lead Developer, team of 5) — Top 6, SDTH — Sep 2026

**Stack:** `Python, WebSockets, WGS84 Geodesic Transforms, Pytest (236 unit tests), HTML5 Canvas Replay Engine`

- Built 6-sensor output pipeline simulation (Radar, Acoustic, RF, IFF transponder, EO, OSINT) across 6 independent clock domains with physical measurement uncertainty bounds and SHA-256 emitter provenance validation.
- Built proof-of-concept operator replay console and THALES WGS84-to-local (ENU) coordinate conversion; team system validated by **236 automated unit tests** under adversarial drone swarm scenarios.

**Aiko** – Autonomous Personal OS Agent Runtime (Solo Developer) — Apr 2026 – Present

**Stack:** `TypeScript, Node.js, SQLite, React Native, Tailscale, Ollama, OpenRouter, Qwen3-TTS`

- Designed 5-tier persistent agent memory architecture with hybrid BM25 and vector embeddings unified via Reciprocal Rank Fusion (RRF); **reduced prompt token payload by 69%** (109 KB to 34 KB) while preserving long-term temporal grounding.
- Engineered autonomous multi-model runtime with automatic provider failover, process watchdog supervision, self-healing diagnostic loops, local GPU Qwen3-TTS voice synthesis, and sandboxed event execution.

**Agora** – Collaborative Multi-Agent IDE & Dev Environment (Lead Developer, team of 3) — Jun 2026 – Oct 2026

SUP Buildathon

**Stack:** `TypeScript, VS Code Core Fork, Mattermost Mod, WebSockets, Node.js, Docker`

- Led 3-man development team; architected parallel AI coding agent orchestrator with AST-level Git 3-way merge conflict resolution.
- Constructed custom Mattermost extension, CLI/API agent harness, and WebSocket state synchronization protocol coordinating concurrent agent modifications within containerized workspaces.

**Kampung Kaki** – Community Emergency Response & Aid Platform (Solo Developer) — May 2026 – Jun 2026

Open Finalist, CODE_EXP

**Stack:** `Python, FastAPI, MQTT 5 QoS 1, Redis, MapLibre GL, OneMap SG GIS, React, Docker`

- Architected distributed event-driven emergency dispatch system; every incident travels as a retained MQTT 5 QoS 1 message with Redis as the shared state, so citizens and first responders, late joiners included, see the same picture.
- Integrated Singapore OneMap GIS APIs and MapLibre GL vector tiles for dynamic role-segregated tactical operation views.

**AnAn** – Elderly Care Fall & Inactivity Monitoring Companion (Solo Developer) — Aug 2026

Tencent Cloud Age Well Social Good Challenge

**Stack:** `Python, FastAPI, MediaPipe, WebAssembly (WASM), Nemotron, Qwen3-TTS, SQLite`

- Constructed 7-stage deterministic safety kernel with state revalidation and permission gating prior to automated emergency alerts.
- Built client-side browser computer vision using MediaPipe/WASM for fall and inactivity detection—guaranteeing **0 raw video frames** leave the user's local device for absolute privacy compliance.

## TECHNICAL SKILLS & CERTIFICATIONS

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|---|---|
| **Languages:** | Python, C/C++, TypeScript, JavaScript, Java, SQL, Rust, Bash, HTML/CSS |
| **AI & Edge Perception:** | TensorRT, PyTorch, SAM3, YOLO11, MediaPipe (WASM), Hybrid Retrieval (BM25 + Vector RRF), Local LLMs (Ollama), Qwen3-TTS |
| **Robotics & Systems:** | ROS 2 (Jazzy), Nav2 MPPI, LiDAR SLAM, Jetson Orin AGX, CAN bus, Docker, Linux, MQTT 5, Redis, WebSockets, MWAN3 |
| **Certifications:** | Google AI Professional Certificate (2026) • Deep Learning Engineering Specialization (Coursera, 2026) |